Vendor Performance Analysis

1. Project Overview

The objective of this project was to analyse over **12+ Million records** present as retail inventory, purchases, and sales data in order to identify business performance trends, vendor performance, and product profitability.

The analysis focuses on understanding:

- Which **brands and vendors drive the most sales**
- Which **vendors and brands underperform**
- Overall **profitability and inventory efficiency**
- Capital locked in unsold inventory

The final insights were visualized using **Microsoft Power BI** to support business decision-making.

2.Data Sources

The dataset contained **over 12 million records** across multiple tables .

```
[7]: print(df.shape)
(12825363, 14)
[8]: df.tail()
```

[8]:

	InventoryId	Store	Brand	Description	Size	SalesQuantity	SalesDollars	SalesPrice	SalesDate	Volume	Classification	ExciseTax	VendorNo	Ver
12825358	9_BLACKPOOL_984	9	984	Patron XO Cafe Liqueur	750mL	1	25.99	25.99	2024-12-17	750.0	1	0.79	9165	
12825359	9_BLACKPOOL_984	9	984	Patron XO Cafe Liqueur	750mL	1	25.99	25.99	2024-12-21	750.0	1	0.79	9165	
12825360	9_BLACKPOOL_984	9	984	Patron XO Cafe Liqueur	750mL	3	77.97	25.99	2024-12-23	750.0	1	2.36	9165	
12825361	9_BLACKPOOL_984	9	984	Patron XO Cafe Liqueur	750mL	1	25.99	25.99	2024-12-24	750.0	1	0.79	9165	
12825362	9_BLACKPOOL_984	9	984	Patron XO Cafe Liqueur	750mL	2	51.98	25.99	2024-12-29	750.0	1	1.57	9165	

TABLES USED:-

begin_inventory

Data Output Messages Notifications

Showing rows: 1 to 5 Page No: 1 of 1

	InventoryId text	Store bigint	City text	Brand bigint	Description text	Size text	onHand bigint	Price double precision	startDate text
1	1_HARDERSFIELD_...	1	HARDERSFIE...	58	Gekkeikan Black & Gold Sa...	750...	8	12.99	2024-01-01
2	1_HARDERSFIELD_...	1	HARDERSFIE...	60	Canadian Club 1858 VAP	750...	7	10.99	2024-01-01
3	1_HARDERSFIELD_...	1	HARDERSFIE...	62	Herradura Silver Tequila	750...	6	36.99	2024-01-01
4	1_HARDERSFIELD_...	1	HARDERSFIE...	63	Herradura Reposado Tequi...	750...	3	38.99	2024-01-01
5	1_HARDERSFIELD_...	1	HARDERSFIE...	72	No. 3 London Dry Gin	750...	6	34.99	2024-01-01

end_inventory

Data Output Messages Notifications

Showing rows: 1 to 5 Page No: 1

	InventoryId text	Store bigint	City text	Brand bigint	Description text	Size text	onHand bigint	Price double precision	endDate text
1	1_HARDERSFIELD_...	1	HARDERSFIE...	58	Gekkeikan Black & Gold Sa...	750...	11	12.99	2024-12-...
2	1_HARDERSFIELD_...	1	HARDERSFIE...	62	Herradura Silver Tequila	750...	7	36.99	2024-12-...
3	1_HARDERSFIELD_...	1	HARDERSFIE...	63	Herradura Reposado Tequi...	750...	7	38.99	2024-12-...
4	1_HARDERSFIELD_...	1	HARDERSFIE...	72	No. 3 London Dry Gin	750...	4	34.99	2024-12-...
5	1_HARDERSFIELD_...	1	HARDERSFIE...	75	Three Olives Tomato Vodka	750...	7	14.99	2024-12-...

purchase_prices

Data Output Messages Notifications

Showing rows: 1 to 5 Page No: 1 of 1

	Brand bigint	Description text	Price double precision	Size text	Volume text	Classification bigint	PurchasePrice double precision	VendorNumber bigint	VendorName text
1	58	Gekkeikan Black & Gold Sa...	12.99	750...	750	1	9.28	8320	SHAW ROSS INT L IMP L...
2	62	Herradura Silver Tequila	36.99	750...	750	1	28.67	1128	BROWN-FORMAN CORP ...
3	63	Herradura Reposado Tequi...	38.99	750...	750	1	30.46	1128	BROWN-FORMAN CORP ...
4	72	No. 3 London Dry Gin	34.99	750...	750	1	26.11	9165	ULTRA BEVERAGE COMP...
5	75	Three Olives Tomato Vodka	14.99	750...	750	1	10.94	7245	PROXIMO SPIRITS INC. ...

Purchases

Data Output Messages Notifications

Showing rows: 1 to 5 Page No: 1 of 1

	InventoryId text	Store bigint	Brand bigint	Description text	Size text	VendorNumber bigint	VendorName text	PONumber bigint	PODate text	ReceivingDate text
1	69_MOUNTMEND_84...	69	8412	Tequila Ocho Plata Fresno	750...	105	ALTAMAR BRANDS LLC	8124	2023-12-...	2024-01-02
2	30_CULCHETH_5255	30	5255	TGI Fridays Ultimte Mudsli...	1.75L	4466	AMERICAN VINTAGE BEVERAGE	8137	2023-12-...	2024-01-01
3	34_PITMERDEN_5215	34	5215	TGI Fridays Long Island Ic...	1.75L	4466	AMERICAN VINTAGE BEVERAGE	8137	2023-12-...	2024-01-02
4	1_HARDERSFIELD_52...	1	5255	TGI Fridays Ultimte Mudsli...	1.75L	4466	AMERICAN VINTAGE BEVERAGE	8137	2023-12-...	2024-01-01
5	76_DONCASTER_2034	76	2034	Glendalough Double Barrel	750...	388	ATLANTIC IMPORTING COMPA...	8169	2023-12-...	2024-01-02

Sales

Data Output Messages Notifications

Showing rows: 1 to 5 Page No: 1 of 1

	InventoryId text	Store bigint	Brand bigint	Description text	Size text	SalesQuantity bigint	SalesDollars double precision	SalesPrice double precision	SalesDate text	Volume double precision	Classific bigint
1	73_DONCASTER_42...	73	4224	Goslings Black Seal Rum 80	750...	3	47.97	15.99	2024-06-28	750	
2	73_DONCASTER_42...	73	4224	Goslings Black Seal Rum 80	750...	2	31.98	15.99	2024-06-29	750	
3	73_DONCASTER_42...	73	4225	Goslings Black Seal Bermu...	1.75L	4	103.96	25.99	2024-06-04	1750	
4	73_DONCASTER_42...	73	4225	Goslings Black Seal Bermu...	1.75L	14	363.86	25.99	2024-06-05	1750	
5	73_DONCASTER_42...	73	4225	Goslings Black Seal Bermu...	1.75L	24	623.76	25.99	2024-06-11	1750	

Vendor_Invoice

Data Output Messages Notifications

Showing rows: 1 to 5 Page No: 1 of 1

	VendorNumber bigint	VendorName text	InvoiceDate text	PONumber bigint	PODate text	PayDate text	Quantity bigint	Dollars double precision	Freight double precision	Approval text
1	105	ALTAMAR BRANDS LLC	2024-01-04	8124	2023-12-...	2024-02-...	6	214.26	3.47	[null]
2	4466	AMERICAN VINTAGE BEVE...	2024-01-07	8137	2023-12-...	2024-02-...	15	140.55	8.57	[null]
3	388	ATLANTIC IMPORTING CO...	2024-01-09	8169	2023-12-...	2024-02-...	5	106.6	4.61	[null]
4	480	BACARDI USA INC	2024-01-12	8106	2023-12-...	2024-02-...	10100	137483.78	2935.2	[null]
5	516	BANFI PRODUCTS CORP	2024-01-07	8170	2023-12-...	2024-02-...	1935	15527.25	429.2	[null]

The project followed a structured data pipeline:

Step 1 — Data Import

CSV files were imported into **Python** using **Pandas**.

Step 2 — Database Storage

All datasets were stored in **PostgreSQL** for efficient querying and handling of large data volumes.

The database contained **12+ million records across six tables**.

Step 3 — Data Aggregation

Using SQL joins and aggregation queries, the datasets were combined to create a **summary table** that consolidated key metrics for analysis.

The final summary table contained **approximately 10,000 rows**, significantly reducing the dataset size while preserving key business metrics.

Raw Data: 12,000,000+ rows

↓

SQL Aggregation

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Final Analysis Dataset: 10,000 rows

Data Output Messages Notifications

Showing rows: 1 to 1000 Page No: 1 of 11

	vendornumber bigint	vendorname text	brand bigint	description text	purchaseprice double precision	actual_price double precision	volume double precision	total_purchase_quantity bigint
1	1128	BROWN-FORMAN CORP	1233	Jack Daniels No 7 Black	26.27	36.99	1750	145080
2	4425	MARTIGNETTI COMPANIES	3405	Tito's Handmade Vodka	23.19	28.99	1750	164038
3	17035	PERNOD RICARD USA	8068	Absolut 80 Proof	18.24	24.99	1750	187407
4	3960	DIAGEO NORTH AMERICA INC	4261	Capt Morgan Spiced Rum	16.17	22.99	1750	201682
5	3960	DIAGEO NORTH AMERICA INC	3545	Ketel One Vodka	21.89	29.99	1750	138109
6	480	BACARDI USA INC	3858	Grey Goose Vodka	17.77	23.99	750	138809
7	17035	PERNOD RICARD USA	2589	Jameson Irish Whiskey	30.76	39.99	1750	70783
8	3960	DIAGEO NORTH AMERICA INC	3102	Smirnoff Traveler	12.94	17.99	1750	161386
9	3960	DIAGEO NORTH AMERICA INC	3489	Tanqueray	20.73	27.99	1750	91835
10	12546	JIM BEAM BRANDS COMPANY	1376	Jim Beam	16.29	21.99	1750	108866
11	17035	PERNOD RICARD USA	2585	Jameson Irish Whiskey	16.42	22.99	750	93205
12	3960	DIAGEO NORTH AMERICA INC	2757	Johnnie Walker Red Label	22.38	29.99	1750	66942
13	480	BACARDI USA INC	2663	Dewars White Label	21.42	29.99	1750	69001
14	10000	MAJESTIC FINE WINES	6570	Kendall Jackson Chard Vt...	6.77	10.99	750	215668
15	9165	ULTRA BEVERAGE COMPANY L...	3650	Patron Silver Tequila	33.81	48.99	750	42951
16	3960	DIAGEO NORTH AMERICA INC	2753	Johnnie Walker Black Label	44.77	59.99	1750	31596
17	17035	PERNOD RICARD USA	5364	Kahlua	23.07	29.99	1750	60645

Total rows: 10692 Query complete 00:00:00.640 CRLF Ln 49, Col 11

Data Cleaning and Feature Engineering

The summary dataset was imported back into **Jupyter Notebook** for exploratory data analysis (EDA).

The following preprocessing steps were performed:

- Handling missing values
- Data type corrections
- Data validation and consistency checks

New calculated metrics

Several analytical features were created to measure business performance:

Metric	Description
Gross Profit	Sales revenue minus purchase cost
Profit Margin	Percentage profit from sales
Stock Turnover	Inventory efficiency metric
Sales to Purchase Ratio	Sales relative to purchased inventory

gross_profit	profit_margin	stock_turnover	sales_to_purchase_ratio
1290667.91	25.30	0.979108	1.338647
1015032.27	21.06	0.976890	1.266830
1119816.92	24.68	0.998575	1.327594
1214774.94	27.14	0.993703	1.372493
1199901.61	28.41	0.983556	1.396897

Exploratory Data Analysis (EDA)

EDA was conducted to understand sales distribution, vendor performance, and brand performance.

Key areas explored included:

- Sales distribution across vendors
- Product brand performance
- Inventory turnover
- Profitability metrics
- Unsold inventory

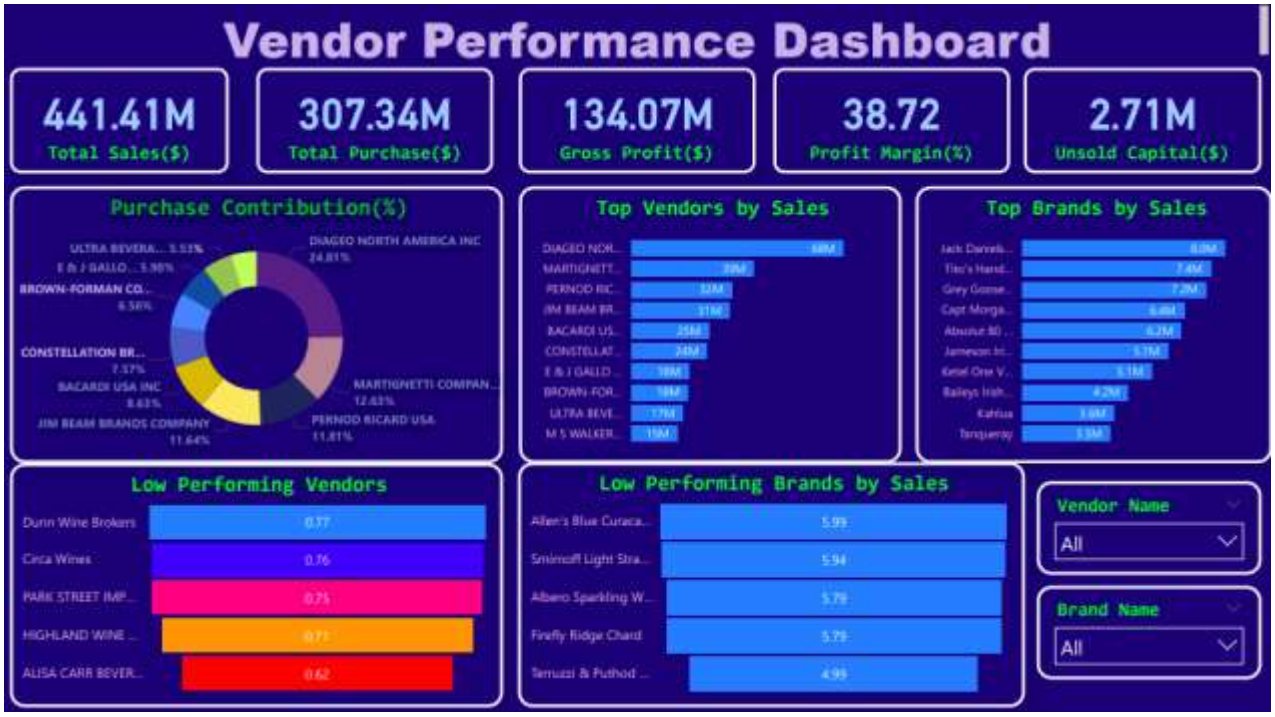
Data Visualization

Interactive dashboards were built using **Microsoft Power BI**.

Key Visualizations

The dashboard includes:

- Top brands by total sales
- Top vendors by total sales
- Low performing vendors
- Low performing brands
- Sales contribution of top 10 brands



Business Recommendations

Based on the analysis, the following actions are recommended:

- Increase focus on **top performing brands and vendors**
- Revaluate relationships with **low performing vendors**
- Reduce purchases from **slow-moving inventory items**
- Improve **inventory turnover efficiency**
- Monitor **unsold capital** to reduce financial risk